

Mathematics for Biomedical Engineering

L3:

Vectors

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Key Learning Objectives

- to understand the basic mathematical representation of a vector and its geometric interpretation
- to perform dot and cross product evaluations with vectors in 3-dimensions
- to understand the mathematical and geometrical representation of line and planes in 3-dimensional space
- to be able to solve complex geometrical problems involving line and planes

- 1 Basic Concepts of Vectors
- 2 Cartesian Components of Vectors
- 3 The scalar or dot product
- 4 The Vector or Cross Product
- 5 The Vector Equation of a Line and a Plane

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Introduction to vectors

Scalar quantities have only a magnitude (or size). Examples include mass, temperature.

Vector quantities have both a magnitude *and direction*. Examples include velocity, force. Vectors can be manipulated according to particular, and well-defined rules.

Quantity	Type	SI unit
Distance	Scalar	metre, m
Area	Scalar	metres squared, m ²
Mass	Scalar	kilogram, kg
Temperature	Scalar	kelvin, K
Pressure	Scalar	pascal, Pa
Work	Scalar	joule, J
Energy	Scalar	joule, J
Displacement	Vector	metre, m
Force	Vector	newton, N
Velocity	Vector	metres per second, m s ⁻¹
Acceleration	Vector	metres per second per second, m s ⁻²

Figure: Examples of some vectors and scalars.

Mathematical description of vector quantities

A vector is often represented as a **directed line segment**:

- the length of the line represents the magnitude of the vector (or its modulus);
- the direction of the line represents the direction of the vector.

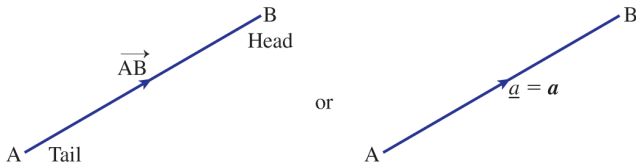


Figure: Representations of vectors. A is the tail, B is the head of the vector.

Vector representations

Vectors are written in different ways:

$$\vec{AB} \qquad \mathbf{a} \qquad \underline{a} \qquad \bar{a}$$

are all ways of representing the vector from A to B above.

[Note: this is VERY important to distinguish vectors from scalars, and can be difficult when writing by hand.]

Mathematical description of vector quantities

Vector magnitude representations

The magnitude of a vector can also be written in different ways:

$$|\vec{AB}| \quad |\mathbf{a}| \quad |\underline{a}| \quad |\bar{a}| \quad a$$

are all ways of representing the magnitude or **modulus** of a vector

Two vectors are said to be equal if they have the same magnitude and direction (even if their locations differ in space). This property allows vectors to be translated without changing the vector itself. These vectors are called **free vectors**.

An *exception* is where a vector is used to represent the position of an object, located at a particular point in space. Such a vector cannot be translated freely.

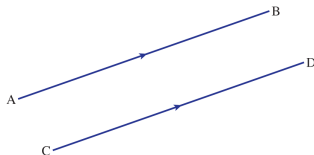


Figure: Two equal vectors.

Mathematical description of vector quantities

The vector $-\mathbf{a}$ is a vector in the opposite direction to $\mathbf{a}$, but with the same magnitude as $\mathbf{a}$.

Geometrically, if $\mathbf{a} = \overrightarrow{AB}$ then $-\mathbf{a} = \overrightarrow{BA}$.

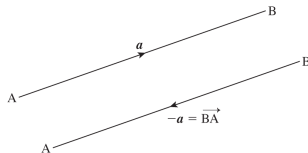


Figure: The negative vector $-\mathbf{a}$ has the opposite direction to the vector $\mathbf{a}$.

Addition and subtraction of vectors

Vectors are added together using the **triangle law**.

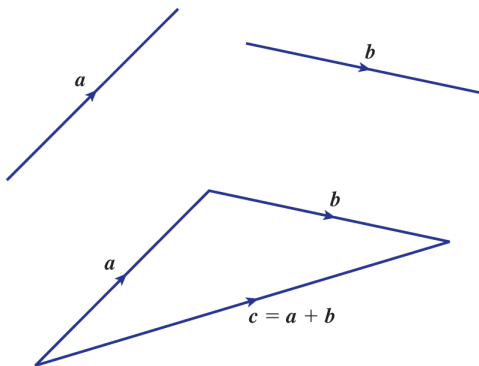


Figure: Addition of two vectors using the triangle law.

Here, vector $\mathbf{b}$ is translated (keeping its length and direction unchanged) so that its tail coincides with the head of $\mathbf{a}$.

The sum of $\mathbf{a} + \mathbf{b}$ is represented by the third side of the triangle.

Addition and subtraction of vectors

Resultant vectors

The **resultant** of two vectors **a** and **b** is their **vector sum** $\mathbf{a} + \mathbf{b}$.

This is useful for finding the combined effect of two forces, known as the **resultant force**.

$$\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2$$

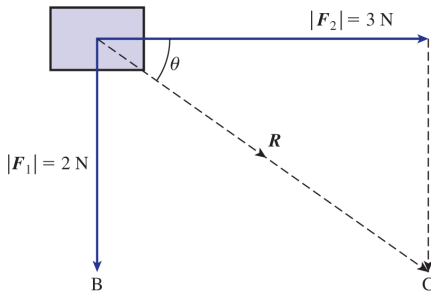


Figure: The combined effect of forces $\mathbf{F}_1$ and $\mathbf{F}_2$ is the resultant $\mathbf{R}$.

Addition and subtraction of vectors

We've just seen that two forces acting on a body may be replaced by a single force (the resultant) which has the same effect.

Sometimes it is useful to consider a single force as two forces acting at right-angles to each other. This is called **resolving** the single force into two (perpendicular) *components*.

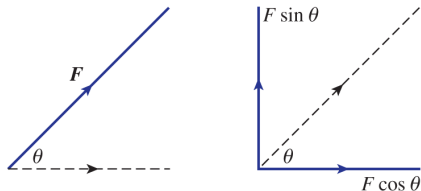


Figure: $\mathbf{F}$ has been resolved into two perpendicular components.

Example: A force of 15 N acts at 40 degrees to the x-axis. Resolve this force into its x- and y-components.

Addition and subtraction of vectors

Subtraction of one vector from another is simply performed by adding the corresponding negative vector i.e. $\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b})$.

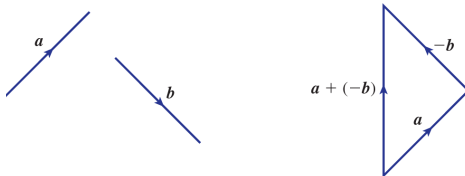


Figure: Vector subtraction.

Geometrical significance of vector subtraction

Note that given two vectors $\mathbf{p} = \vec{OP}$ and $\mathbf{q} = \vec{OQ}$, the vector from P to Q is given by $\mathbf{q} - \mathbf{p}$.

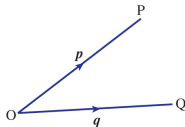


Figure: Vector subtraction.

Multiplication by a scalar

If k is any positive scalar (number) and $\mathbf{a}$ is a vector, then $k\mathbf{a}$ is a vector:

- in the *same direction* as $\mathbf{a}$;
- with a magnitude (length) that is k times as long as $\mathbf{a}$;
- if k is negative, $k\mathbf{a}$ is in the opposite direction to $\mathbf{a}$ (and still k times as long).

The vector $k\mathbf{a}$ is said to be a **scalar multiple** of $\mathbf{a}$.



Figure: Multiplying a vector by a scalar.

Rules of scalar multiplication

$$k(\mathbf{a} + \mathbf{b}) = k\mathbf{a} + k\mathbf{b} \quad (1)$$

$$(k + l)\mathbf{a} = k\mathbf{a} + l\mathbf{a} \quad (2)$$

$$k(l\mathbf{a}) = (kl)\mathbf{a} \quad (3)$$

Unit vectors

A unit vector:

- is a vector with a modulus of 1;
- represents the direction of a vector only.

To obtain the unit vector in the direction of any vector $\mathbf{a}$, we divide the vector $\mathbf{a}$ by its modulus.

Unit vector calculation

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|} \quad (4)$$

where a unit vector is represented by a 'hat' on top of the vector.

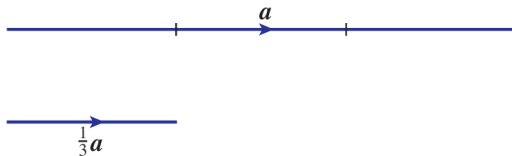


Figure: If vector $\mathbf{a}$ has a modulus of 3, the unit vector in the direction of $\mathbf{a}$ is $\frac{1}{3}\mathbf{a}$.

Example of basic concepts

Example: In the triangle OAB , the point P exactly divides side AB at the mid-point. If $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$:

- *Depict this on a diagram.*
- *Find an expression for $\vec{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.*
- *What is the unit vector $\vec{OP}$?*

- 1 Basic Concepts of Vectors
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Two-dimensional coordinate frames and the vectors $\mathbf{i}$ and $\mathbf{j}$

Any vector in the xy -plane can be represented as a combination of *unit vectors* in the direction of the x - and y -axes - called **basis vectors**.

The unit vectors $\mathbf{i}$ and $\mathbf{j}$

A unit vector pointing in the positive x -direction is denoted $\mathbf{i}$.

A unit vector pointing in the positive y -direction is denoted $\mathbf{j}$.

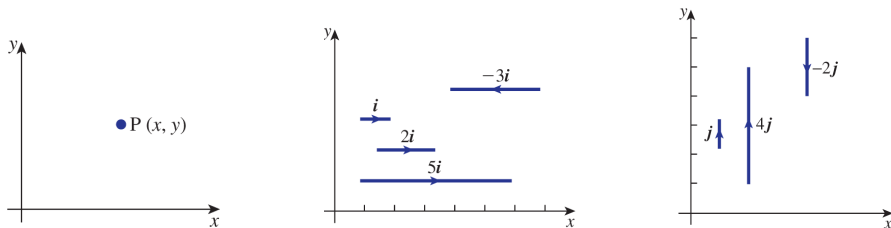


Figure: A 2D coordinate frame (left) with all vectors multiples of $\mathbf{i}$ (centre) and all vectors multiples of $\mathbf{j}$ (right)

Any vector in the x -direction will be a multiple of $\mathbf{i}$; similarly, any vector in the y -direction will be a multiple of $\mathbf{j}$.

Two-dimensional coordinate frames and the vectors **i** and **j**

We can represent any vector **r** as the resultant of two vectors in the *x*- and *y*-directions.

$$\mathbf{r} = \vec{AB} = \vec{AC} + \vec{CB} = a\mathbf{i} + b\mathbf{j} \quad (5)$$

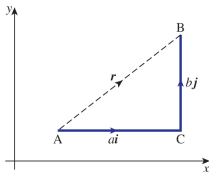


Figure: Vector **r** in terms of **i** and **j**.

r vector

Any vector in the *xy*-plane can be written as

$$\mathbf{r} = a\mathbf{i} + b\mathbf{j} = \begin{pmatrix} a \\ b \end{pmatrix} \quad (6)$$

The components *a* and *b* are called the **i** and **j** **components** of **r**.

Two-dimensional coordinate frames and the vectors $\mathbf{i}$ and $\mathbf{j}$

Vectors in cartesian form can be added by simply adding their respective $\mathbf{i}$ and $\mathbf{j}$ components.

Addition and subtraction of vectors

If $\mathbf{a} = a_x\mathbf{i} + a_y\mathbf{j}$ and $\mathbf{b} = b_x\mathbf{i} + b_y\mathbf{j}$ then

$$\mathbf{a} + \mathbf{b} = (a_x + b_x)\mathbf{i} + (a_y + b_y)\mathbf{j} \quad (7)$$

Example: If $\mathbf{a} = 9\mathbf{i} + 7\mathbf{j}$ and $\mathbf{b} = 8\mathbf{i} + 3\mathbf{j}$ find (a) $\mathbf{a} + \mathbf{b}$ and (b) $\mathbf{a} - \mathbf{b}$.

Scalar multiplication of vectors

If $\mathbf{a} = a_x\mathbf{i} + a_y\mathbf{j}$ then

$$k\mathbf{a} = ka_x\mathbf{i} + ka_y\mathbf{j} \quad (8)$$

Example: If $\mathbf{a} = 9\mathbf{i} + 3\mathbf{j}$ write down (a) $4\mathbf{a}$ and (b) $-\frac{1}{3}\mathbf{a}$.

Two-dimensional coordinate frames and the vectors $\mathbf{i}$ and $\mathbf{j}$

Position vector

The **position vector** of a point $P(x, y)$ from the origin is

$$\mathbf{r} = \vec{OP} = x\mathbf{i} + y\mathbf{j} \quad (9)$$

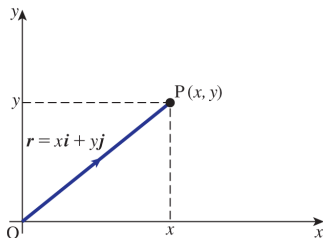


Figure: Position vector of the point P with coordinates (x, y) .

Position vectors cannot be freely translated. Since they indicate the position of a point, they are fixed vectors in the sense that the tail of a position vector is always located at the origin.

Two-dimensional coordinate frames and the vectors $\mathbf{i}$ and $\mathbf{j}$

The **modulus** of any vector $\mathbf{r}$ is equal to its length, which may simply be obtained by Pythagoras's Theorem.

Modulus

If $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$ then

$$|\mathbf{r}| = \sqrt{x^2 + y^2} \quad (10)$$

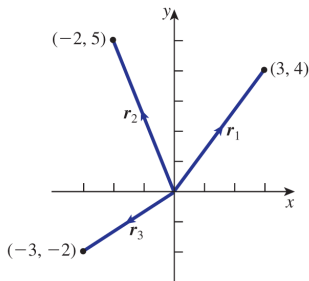


Figure: Position vectors of points with coordinates $(3, 4)$, $(-2, 5)$, $(-3, -2)$.

The direction ratios and direction cosines

Consider the point $P(4, 5)$ and its position vector $4\mathbf{i} + 5\mathbf{j}$.

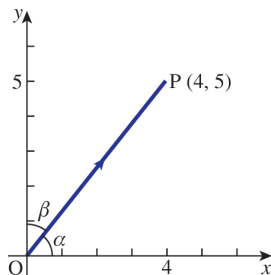


Figure: The direction cosines of $\vec{OP}$ are $\cos \alpha$ and $\cos \beta$.

The **direction ratio** of $\vec{OP}$ is 4 : 5.

The **direction cosines** of $\vec{OP}$ are the cosines of the angles between the vector and each of the positive axes (i.e. the positive x- and y-axes).

$$\cos \alpha = \frac{4}{\sqrt{41}} \qquad \cos \beta = \frac{5}{\sqrt{41}} \qquad (11)$$

The direction ratios and direction cosines

The cosine ratios are conventionally labelled l and m .

Cosine ratios

For any vector $\mathbf{r} = a\mathbf{i} + b\mathbf{j}$, the **cosine ratios** are

$$l = \frac{a}{\sqrt{a^2 + b^2}} \quad m = \frac{b}{\sqrt{a^2 + b^2}} \quad (12)$$

Note that the summation of the square of the direction cosines equals 1

$$l^2 + m^2 = 1 \quad (13)$$

Three-dimensional coordinate frames

In three-dimensions, the x -, y - and z -directions are **mutually perpendicular**.

Any point P in three-dimensional space can be defined by its x -, y - and z -coordinates.

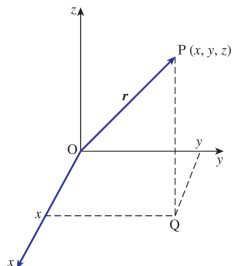


Figure: Point in 3D space with coordinates (x, y, z) with position vector $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$

The unit vector $\mathbf{k}$ is used to denote the positive z -direction.

Position vector in 3D

The position vector $P(x, y, z)$ is

$$\mathbf{r} = \vec{OP} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \quad (14)$$

Direction ratios and direction cosines in three dimensions

The concepts of direction ratios and cosines extend naturally to 3D.

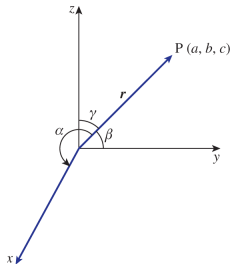


Figure: The direction cosines of $\vec{OP}$ are $\cos \alpha, \cos \beta, \cos \gamma$.

Cosine ratios

For any vector $\mathbf{r} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$ the direction cosines are

$$l = \frac{a}{\sqrt{a^2 + b^2 + c^2}}, \quad m = \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \quad n = \frac{c}{\sqrt{a^2 + b^2 + c^2}} \quad (15)$$

and

$$l^2 + m^2 + n^2 = 1 \quad (16)$$

N-dimensional vectors

Vectors are often used to represent quantities in engineering in higher dimensions. No geometric interpretation is then available. Nevertheless, it is still possible to represent and manipulate vectors of higher dimensions using exactly the same rules as learnt for lower dimensions.

For higher dimension vectors, the modulus of the vector is usually referred to as its **norm**.

In engineering, signals are often represented as vectors containing amplitude values at a series of time-points

$$\mathbf{f} = (f_0, f_1, f_2, \dots, f_{n-1}) \quad (17)$$

We can find a measure of the difference between this signal and another signal $\mathbf{f}'$ by taking $|\mathbf{f} - \mathbf{f}'|$.

- 1 Basic Concepts of Vectors
- 2 Cartesian Components of Vectors
- 3 The scalar or dot product
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Definition of the scalar product

The **scalar product** (or **dot product**) is a way of multiplying two vectors together.

Scalar Product

The scalar product of two vectors **a** and **b** is given by

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta, \quad (18)$$

where θ is the angle between the two vectors; *length of first vector, times length of second vector, times the cosine of the angle between them.*

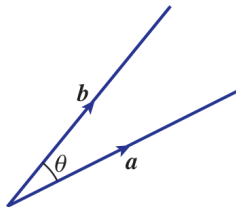


Figure: Two vectors, separated by angle θ .

Definition of the scalar product

From the definition of the scalar product, we can see that

$$\mathbf{i} \cdot \mathbf{i} = \mathbf{j} \cdot \mathbf{j} = \mathbf{k} \cdot \mathbf{k} = 1 \quad (19)$$

$$\mathbf{i} \cdot \mathbf{j} = \mathbf{i} \cdot \mathbf{k} = \mathbf{j} \cdot \mathbf{k} = 0 \quad (20)$$

$$(21)$$

Example: Vectors $\mathbf{a}$ and $\mathbf{b}$ has moduli 6 and 7, respectively. The angle between the vectors is 60 degree. Compute $\mathbf{a} \cdot \mathbf{b}$.

Example: A force of 15 N acts in the x -direction. What is the component of the force in the y -direction and why?

Properties of the scalar product

Commutativity

Since the RHS of the scalar product definition just involves numbers (which commute), it follows that

$$|\mathbf{a}| |\mathbf{b}| \cos \theta = |\mathbf{b}| |\mathbf{a}| \cos \theta \quad (22)$$

and so

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a} \quad (23)$$

We can evaluate the scalar product in any order!

Distributivity For 3 vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$

$$\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c} \quad (24)$$

The scalar product is distributive over addition.

Multiplication by a scalar

For two vectors $\mathbf{a}$ and $\mathbf{b}$, and a scalar k ,

$$k(\mathbf{a} \cdot \mathbf{b}) = (k\mathbf{a}) \cdot \mathbf{b} = \mathbf{a} \cdot (k\mathbf{b}) \quad (25)$$

Properties of the scalar product

Example: Simplify $\mathbf{i} \cdot 7\mathbf{i}$.

Example: Find $\mathbf{i} \cdot (\mathbf{i} + \mathbf{j})$.

Example: Simplify $(3\mathbf{i} + 2\mathbf{j}) \cdot (7\mathbf{i} + 4\mathbf{j})$.

A formula for finding the scalar product

Scalar product formula

If $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j}$, then

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2. \quad (26)$$

Above we have used that $\mathbf{i} \cdot \mathbf{i} = \mathbf{j} \cdot \mathbf{j} = 1$ and $\mathbf{i} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{i} = 0$.

Thus, to compute the scalar product of two vectors, their individual i and j components are multiplied together and added-up.

Example: If $\mathbf{p} = 5\mathbf{i} - 3\mathbf{j}$ and $\mathbf{q} = 2\mathbf{i} + \mathbf{j}$, find $\mathbf{p} \cdot \mathbf{q}$.

Scalar product formula in 3D

In 3D, if $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$, the formula is simply extended to

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 \quad (27)$$

Example: If $\mathbf{r} = 3\mathbf{i} + 2\mathbf{j} + 9\mathbf{k}$, find $\mathbf{r} \cdot \mathbf{r}$.

Using the scalar product to find the angle between two vectors

Angle between two vectors

The equation for the scalar product can be simply re-arranged to find the angle between two vectors

$$\theta = \cos^{-1} \left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} \right) \quad (28)$$

Example: Find the angle between the two vectors $\mathbf{a} = 5\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$ and $\mathbf{b} = 8\mathbf{i} - 9\mathbf{j} + 11\mathbf{k}$

Using the scalar product to find the component of a vector in the direction of another vector

We can use the vector product to *project* one vector in the direction of another.

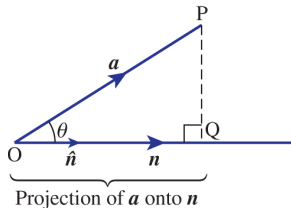


Figure: Projection of vector $\mathbf{a}$ along direction $\hat{\mathbf{n}}$.

The distance OQ is the projection of $\mathbf{a}$ onto $\hat{\mathbf{n}}$, or the component of $\mathbf{a}$ in the direction of $\hat{\mathbf{n}}$.

The length OQ is obviously (from trigonometry) equal to $|\mathbf{a}| \cos \theta$.

Using the scalar product to find the component of a vector in the direction of another vector

However, we know that taking the scalar product of $\mathbf{a}$ with $\hat{\mathbf{n}}$ gives

$$\mathbf{a} \cdot \hat{\mathbf{n}} = |\mathbf{a}| |\hat{\mathbf{n}}| \cos \theta \quad (29)$$

$$= |\mathbf{a}| \cos \theta \quad (\text{since } |\hat{\mathbf{n}}| = 1) \quad (30)$$

which is just the length of OQ.

Projection of a vector

$\mathbf{a} \cdot \hat{\mathbf{n}}$ is the length of the projection of $\mathbf{a}$ onto $\hat{\mathbf{n}}$

$\mathbf{a} \cdot \hat{\mathbf{n}}$ is the component of $\mathbf{a}$ in the direction of $\hat{\mathbf{n}}$

Example: Find the component of the vector $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j}$ in the direction $\mathbf{n} = \mathbf{i} + 5\mathbf{j}$.

Application to planes

Consider a plane containing a point A with position vector $\mathbf{a}$. The normal (or perpendicular) to the plane is $\mathbf{n}$. Find an expression for the perpendicular distance of the plane from the origin.

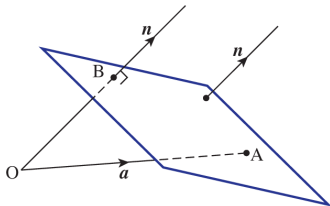


Figure: OB is the perpendicular distance of the plane from the origin.

The perpendicular distance of the plane from the origin is the projection (or component) of $\mathbf{a}$ in the direction of $\hat{\mathbf{n}}$.

- 1 Basic Concepts of Vectors
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The right-hand screw rule

When multiplying two vectors (**a** and **b**) with the **vector (or cross) product** the result is *always a vector*.

Moreover, it is *perpendicular* to *both* vector **a** and **b** - i.e. perpendicular to the plane containing both **a** and **b**.

A right-handed screw enters into a material when screwed in a clockwise direction.

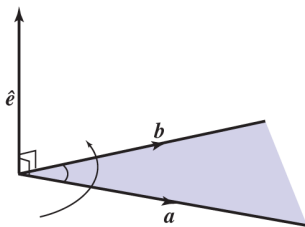


Figure: When turning a right-handed screw from **a** towards **b**, it will advance in the direction given by $\hat{e}$

Definition of the vector product

Vector Product

The **vector product** of the two vectors **a** and **b** is given by the product of the magnitude of each of vectors, multiplied by the sine of the angle between them in the direction $\hat{e}$

$$\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin \theta \hat{e}, \quad (31)$$

where $\hat{e}$ is a unit vector perpendicular to the plane containing **a** and **b** in a sense given by the right-handed screw rule.

The vector product is sometimes called the **cross product**.

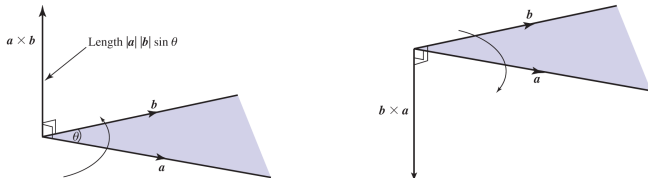


Figure: Value of $\mathbf{a} \times \mathbf{b}$ (left) and $\mathbf{b} \times \mathbf{a}$ (right).

Note that the direction of $\mathbf{a} \times \mathbf{b}$ is *opposite* to $\mathbf{b} \times \mathbf{a}$ i.e. $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$

Definition of the vector product

Note that if $\mathbf{a}$ and $\mathbf{b}$ are *parallel*, the result is the zero vector $\mathbf{a} \times \mathbf{b} = \mathbf{0}$.

Important result (1)

$$\mathbf{i} \times \mathbf{i} = \mathbf{0}$$

$$\mathbf{j} \times \mathbf{j} = \mathbf{0}$$

$$\mathbf{k} \times \mathbf{k} = \mathbf{0}$$

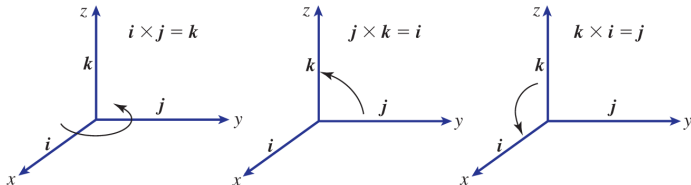
$$(32)$$


Figure: $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are all *mutually perpendicular* (or *orthogonal*) with a modulus of 1.

Important result (2)

$$\mathbf{i} \times \mathbf{j} = \mathbf{k}$$

$$\mathbf{j} \times \mathbf{k} = \mathbf{i}$$

$$\mathbf{k} \times \mathbf{i} = \mathbf{j}$$

$$(33)$$

$$\mathbf{j} \times \mathbf{i} = -\mathbf{k}$$

$$\mathbf{k} \times \mathbf{j} = -\mathbf{i}$$

$$\mathbf{i} \times \mathbf{k} = -\mathbf{j}$$

$$(34)$$

Note the *cyclical permutation* of i, j, k

Some properties of the vector product

Non-commutativity

$$\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a} \quad (35)$$

Distributivity

$$\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c} \quad (36)$$

The vector product is distributive over addition.

Multiplication by a scalar

$$k(\mathbf{a} \times \mathbf{b}) = (k\mathbf{a}) \times \mathbf{b} = \mathbf{a} \times (k\mathbf{b}) \quad (37)$$

A formula for finding the vector product

Suppose

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k} \qquad \mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k} \qquad (38)$$

then by using the previous relationships we get

Formula for the vector product

$$\mathbf{a} \times \mathbf{b} = (a_2b_3 - a_3b_2)\mathbf{i} - (a_1b_3 - a_3b_1)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k} \qquad (39)$$

Example: Simplify $\mathbf{i} \times (\mathbf{i} + \mathbf{j})$.

Example: Find the vector product of $\mathbf{a} = 3\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$ with $\mathbf{b} = 7\mathbf{i} + 4\mathbf{j} - 8\mathbf{k}$.

Using determinants to evaluate a vector product

The vector product of two vectors $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ is also (more easily) found by evaluating the determinant:

Vector product using determinant

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2b_3 - a_3b_2)\mathbf{i} - (a_1b_3 - a_3b_1)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k} \quad (40)$$

where this is evaluated by computing the sub-determinants.

Example: Find the vector product of $\mathbf{a} = 3\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$ with $\mathbf{b} = 9\mathbf{i} - 6\mathbf{j} + 2\mathbf{k}$.

Example of vector product

A force $\mathbf{F}$ applied to an object causes it to rotate around a fixed axis

$$\mathbf{M}_0 = \mathbf{r} \times \mathbf{F} \quad (41)$$

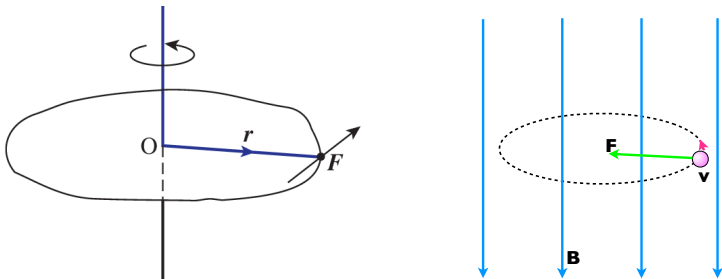


Figure: (left) Moment of a force about an axis. (right) Force on particle moving through magnetic field.

The force experienced by a charged particle (charge q) moving with a velocity $\mathbf{v}$ through a magnetic field $\mathbf{B}$

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B} \quad (42)$$

- 1 Basic Concepts of Vectors
- 2 Cartesian Components of Vectors
- 3 The scalar or dot product
- 4 The Vector or Cross Product
- 5 The Vector Equation of a Line and a Plane

Vector equation of a line

Points A and B are fixed points on the line with position vectors $\mathbf{a}$ and $\mathbf{b}$, respectively.

Point P is any other arbitrary (variable) point on the line with position vector $\mathbf{r}$.

The line $\overrightarrow{AB}$ and $\overrightarrow{AP}$ are parallel.

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} \quad (43)$$

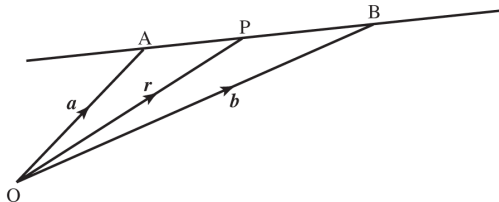


Figure: A straight line passing through points A and B.

Vector equation of a line

$$\mathbf{r} = \mathbf{a} + k(\mathbf{b} - \mathbf{a}) \quad (44)$$

represents the line passing through the points with position vectors $\mathbf{a}$ and $\mathbf{b}$.

Vector equation of a line

$$\mathbf{r} = \mathbf{a} + k(\mathbf{b} - \mathbf{a}) \quad (45)$$

Note that:

- by varying k we move along the line: $k = 0$ means we are at A; $k = 1$ means we are at B etc.
- the vector $(\mathbf{b} - \mathbf{a})$ gives the direction of the line.

Example: Find the vector equation of the line passing through $A(3, 2)$ and $B(7, 5)$.

Example: Do the same for $A(5, -2, 3)$ and $B(2, 1, -4)$.

Cartesian form of equation of a line

Using

$$\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}, \quad \mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad (46)$$

along with

$$\mathbf{r} = \mathbf{a} + k(\mathbf{b} - \mathbf{a}) \quad (47)$$

we can equate individual components and solve for k in each case to get

Cartesian form of equation of a line

$$\frac{x - a_1}{b_1 - a_1} = \frac{y - a_2}{b_2 - a_2} = \frac{z - a_3}{b_3 - a_3} \quad (48)$$

passing through points the points with position vectors (a_1, a_2, a_3) , (b_1, b_2, b_3) .

Note that the denominators above give the components of the direction of the line.

Example: Write down the Cartesian and vector equation for the line passing through $(9, 3, -2)$, $(4, 5, -1)$.

The vector equation of a plane

A is a known point in the plane with position vector $\mathbf{a}$.

P is any other arbitrary point in the plane with position vector $\mathbf{r}$.

$\overrightarrow{AP}$ must lie in the plane and

$$\overrightarrow{AP} = \mathbf{r} - \mathbf{a} \quad (49)$$

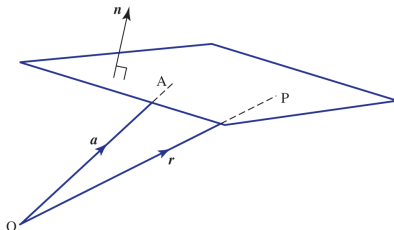


Figure: Plane passes through A and is perpendicular to $\mathbf{n}$.

The vector $\mathbf{n}$ is perpendicular to the plane and is thus *perpendicular* to $\overrightarrow{AP}$

$$(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0 \quad (50)$$

The vector equation of a plane

Vector equation of a plane

A plane passing through the point with position vector $\mathbf{a}$ and being perpendicular to the vector $\mathbf{n}$ is given by

$$\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n} \quad (51)$$

All points within the plane with general position vector $\mathbf{r}$ satisfy this equation.

Note that the Cartesian form of these equations is obtained by letting $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and evaluating the scalar product.

Example: Find the vector equation of the plane passing through the point $(3, 2, 5)$ which is perpendicular to the vector $\mathbf{i} + \mathbf{k}$. Also find the Cartesian equation of this plane.

The vector equation of a plane with unit normal

If $\mathbf{n}$ is a unit vector i.e. $\hat{\mathbf{n}}$, then $\mathbf{a} \cdot \hat{\mathbf{n}}$ represents the *perpendicular distance from the origin to the plane*.

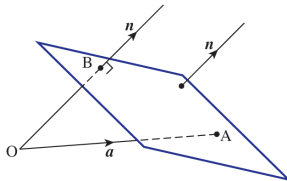


Figure: OB is the perpendicular distance of the plane from the origin.

Vector equation of a plane with unit normal

A plane with unit normal $\hat{\mathbf{n}}$ (containing the point with position vector $\mathbf{a}$) and which is a perpendicular distance d from O is given by

$$\mathbf{r} \cdot \hat{\mathbf{n}} = d \quad (52)$$

where this distance d may be evaluated as

$$d = \mathbf{a} \cdot \hat{\mathbf{n}}. \quad (53)$$

What we learnt today

- to understand the basic mathematical representation of a vector and its geometric interpretation
- to perform dot and cross product evaluations with vectors in 3-dimensions
- to understand the mathematical and geometrical representation of line and planes in 3-dimensional space
- to be able to solve complex geometrical problems involving line and planes